

1. Express Router & Modular Routing

`express.Router()` euta mini-application jastai ho. Yesle farak-farak routing logic lai aafnei file ma separation garna dinchha, jaslai hami modular routing bhanchhaũ. [[1](#), [2](#)]

Real-world Analogy:

Euta thulo Company (App) ma farak-farak Departments (Routers) hunchhan—jastai HR Department (`/auth`), Sales Department (`/products`), ra Support Department (`/tickets`). Sabae kaam central manager le eklai herdaina.

2. Route Organization (Best Folder Structure)

Professional project haruma code yesari organize garienchha:

text

```
my-backend-app/  
├── node_modules/  
├── routes/  
│   ├── authRoutes.js  
│   └── productRoutes.js  
├── index.js  
├── package.json  
└── .env
```

Use code with caution.

3. Modular Routing in Detail (Step-by-Step Code)

Aba hami yo folder structure anusar code kasari link garchhan bhanera herchhaũ.

Step 1: Product Routes Banaune (`routes/productRoutes.js`) [[1](#), [2](#)]

javascript

```
const express = require('express');
```

```
const router = express.Router(); // Mini-router initialize gareko

// Note: Yahan hami purei path lekhdauñ, just '/' matra lekhchhañ
// Main file ma yo router merge garda prefix automatic thapinchha

// 1. GET: Sabai products lina
router.get('/', (req, res) => {
  res.json({ message: "Sabai products ko list yahan chha" });
});

// 2. POST: Naya product thapna
router.post('/', (req, res) => {
  res.json({ message: "Naya product add bhayo" });
});

module.exports = router; // Router lai export gareko
Use code with caution.
```

Step 2: Main Server file ma Link garne (index.js)

javascript

```
const express = require('express');
const app = express();
const PORT = 3000;

// Routers import gareko
const productRoutes = require('./routes/productRoutes');

app.use(express.json());

// Main Magic: Base URL match garesi mathi ko router active hunchha
// Aba /api/products mathi ko file ko '/' sanga connect bhayo
app.use('/api/products', productRoutes);

app.listen(PORT, () => {
  console.log(`Server running on http://localhost:${PORT}`);
});
Use code with caution.
```



4. Route Parameters (`req.params`) vs Query Parameters

(`req.query`)

Yo dubaiko bichma dherai developers झुकिन्छन् (confuse hunchhan). Yo ekdam clear sanga bujhũ:

Feature	Route Parameters (<code>req.params</code>)	Query Parameters (<code>req.query</code>)
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URL Format	/api/products/8848	/api/products?category=shoes&sort=asc
Syntax	URL path ko bhag ho (/ :id)	URL ko end ma ? pachi safe gariyerahanchha (key=value)
Use Case	Specific data direct khojna (Mandatory)	Filter, Search, Pagination, Sort garna (Optional)

Code Implementation Example:

Tapai ko `routes/productRoutes.js` bhitra yo dui tarika yesari use hunchha:

javascript

```
// A. Route Parameter Example
// Request URL: http://localhost:3000/api/products/102
router.get('/:productId', (req, res) => {
    const id = req.params.productId; // '102' linchha
    res.json({ productId: id, status: "Fetched specific product" });
});

// B. Query Parameter Example
// Request URL: http://localhost:3000/api/products?search=laptop&page=2
router.get('/', (req, res) => {
    const searchText = req.query.search; // 'laptop' linchha
    const pageNumber = req.query.page;   // '2' linchha

    res.json({
        searchingFor: searchText,
        page: pageNumber,
        message: "Filtering results based on query"
    });
});
```

Use code with caution.

5. Nested Routes (Sub-routing)

Nested routing bhaneko router bhitra pani arko router nested (khapinuhunu) ho. Jastai: Euta specific product bhitra dherai reviews haru huna sakchhan (`/api/products/:productId/reviews`). [\[1\]](#)

Express ma yesko lagi `mergeParams: true` option use garnuparchha, jasje garda child router le parent router ko parameters read garna sakos.

Product Review Router (`routes/reviewRoutes.js`):

javascript

```
const express = require('express');
```

```
// mergeParams: true le garda mathillo router ko ':productId' yo file le
pani access garna pauchha
const router = express.Router({ mergeParams: true });

// URL: /api/products/:productId/reviews
router.get('/', (req, res) => {
  const pId = req.params.productId; // Parent URL bata aayo
  res.json({ message: `Product ID ${pId} ko sabai reviews haru` });
});

module.exports = router;
Use code with caution.
```

Yo Nested Router lai Product Router ma link garne (`routes/productRoutes.js`):

javascript

```
const express = require('express');
const router = express.Router();
const reviewRoutes = require('./reviewRoutes'); // Review router link
gareko

// Specific product ko reviews endpoint direct forward gareko
router.use('/:productId/reviews', reviewRoutes);

module.exports = router;
Use code with caution.
```

Quick Summary Checklist

- **Express Router** le code structured ra clean banauchha.
- **Route Params (:id)** exact database object fetch garna chahinchha.
- **Query Params (?key=val)** filter, search, ra sorting ko lagi standard ho.
- **mergeParams: true** nested resources handle garna use hunchha. [[1](#), [2](#), [3](#), [4](#), [5](#)]